

“MY... PART IN THE STORY WAS THAT I SAW **SOMETHING UNUSUAL** AND **APPRECIATED** SOMETHING OF **ITS IMPORTANCE** SO... I SET TO WORK ON IT.”

Alexander Fleming, Scottish biologist, address to the University of Edinburgh, 1952

The *Penicillium* mold is a common fungus, spread by spores from the capsules seen here; some species produce the penicillin seen by Fleming.

THE YEAR BEGAN WITH EXPERIMENTAL CONFIRMATION

that characteristics and inheritance were determined by a chemical substance. British physician **Frederick Griffith** (1879–1941) studied strains of pneumonia bacteria, some of which could cause disease, while others were harmless. Griffith's work indicated that a **transforming factor** could move from the harmful bacteria to others and make them harmful too (see panel, below). This factor would later be identified as DNA (see 1943–44).

Another advance came in physics. Austrian physicist Erwin Schrödinger had described the wave function of a particle (see

1926). Now British physicist **Paul Dirac** described a new form of Schrödinger's wave equation for electrons. As a result, he predicted a new class of matter called anti-electrons, which had positive, instead of negative, charges. Dirac's work was the first **modern theory of antimatter**. The anti-electrons would later be discovered and renamed positrons (see 1932–33). It would also become evident that corresponding antimatter particles existed for most subatomic particles.

In September the previous year, Britain's **John Logie Baird** had sent his technical assistant to New York to prepare for the biggest test of his television

system so far: a **transatlantic transmission**. After several false starts, success came at midnight, London time, on February 8, 1928. Baird himself briefly appeared as a fuzzy image on screen in America.

In the wake of World War I, Scottish biologist **Alexander Fleming** (1881–1955) was developing ways of fighting infections that went beyond the routine use of antiseptics, which

were often ineffective for serious wounds. As part of his research, Fleming had been growing cultures of infectious bacteria. On September 3, in his laboratory at St. Mary's Hospital, London, Fleming noticed that one of his cultures had become contaminated: **a mold had spread on the culture dish**. But significantly, just around the spreading mold there was a region that was clear of bacteria.



PAUL DIRAC (1902–84)

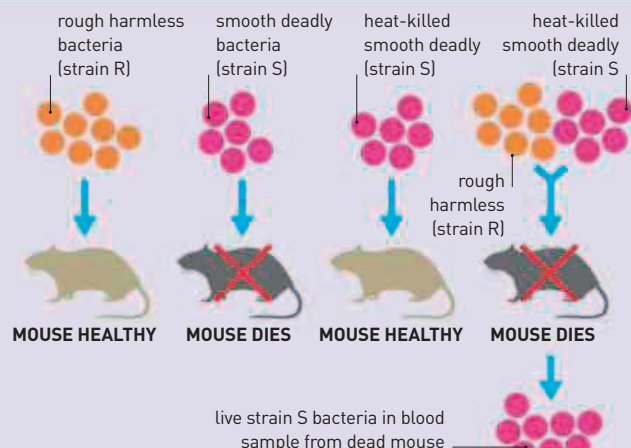
British physicist Paul Dirac held the Lucasian Chair of Mathematics at Cambridge University, UK, from 1932 to 1969. He advanced quantum physics by applying it to Einstein's theory of relativity (see pp.244–45). His work on the quantum wave equations predicted the existence of antimatter. In 1933 he shared the Nobel Prize for Physics with Erwin Schrödinger.

“IF YOU ARE **RECEPTIVE** AND **HUMBLE**, MATHEMATICS **WILL LEAD YOU...**”

Paul Dirac, British physicist, November 27, 1975

BACTERIAL TRANSFORMATION

Pneumococcus bacteria exist in harmless (rough) and harmful (smooth) forms. Frederick Griffith found that bacteria can exchange a chemical that changes these characteristics. Heat-killed harmful bacteria fail to cause an infection on their own, but if mixed with living harmless bacteria, a substance moves from dead bacteria to living ones, changing them into harmful forms that kill the mouse and appear in its blood.



Clearly, the mold was producing something that had **killed the bacteria**. Fleming took samples of the mold, cultivated it, and identified it as *Penicillium*. The following year he called the active ingredient of his **anti-bacterial mold juice penicillin**. Despite his efforts, Fleming was unable to isolate penicillin in its raw chemical form, but he did preserve his culture of mold. In the following decades penicillin would not only be purified then manufactured, but also lauded as one of the **first effective antibiotic cures** of bacterial infection (see 1940–41).

A medical breakthrough of a different kind came in Sydney, Australia. In 1926, physicians **Mark Lidwell** (1878–1969) and **Edgar Booth** (1893–1963) had devised a portable plug-in **artificial heart pacemaker**. In 1928 they used it to **revive a stillborn infant**.

January Frederick Griffith shows that **bacterial transformation** happens because of the transfer of a chemical substance

January 26 Anak Krakatau appears as a new volcanic island in Indonesia

February 8 John Logie Baird sends first transatlantic TV signal

February 1 Paul Dirac predicts the existence of anti-matter

February 28 Indian physicist Chandrasekhara Raman describes the scattering of photons (packages) of light

Mark Lidwell and Edgar Booth use their portable pacemaker to revive a stillborn infant

July 3 John Logie Baird demonstrates first color television

September 3 Alexander Fleming accidentally discovers penicillin at St. Mary's Hospital, London

Late 1928 Hungarian physiologist Albert Szent-Gyorgyi describes isolating a carbohydrate derivative, identified in 1932 as **vitamin C**

Dutch biologist Fritz Went shows that plants use chemical signals, later called **auxins**